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Geochronology and palaeomagnetism of a late Proterozoic island arc terrane from the Red Sea Hills, northeast Sudan

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ABSTRACT

Size fractions and single grains of zircon, hornblende and biotite have been dated to define the formation age and thermal history of the northern Haya Terrane (NHT), a late Proterozoic island arc terrane of the Red Sea Hills (RSH), northeast Sudan. U–Pb and $^{207}Pb/^{206}Pb$ zircon data for acidic volcanics and granitoid rocks bracket the time of formation of the NHT between 887 ± 20 and 854 ± 9 Ma. $^{40}Ar/^{39}Ar$ hornblende and biotite ages of a gabbro near Suakin suggest passage through the argon closing temperature for hornblende at about 841 ± 4 Ma and 782 ± 6 Ma for biotite. The hornblende age is considered to be representative of the age of magnetic blocking and, hence, the age of the characteristic magnetization of the Suakin Gabbro. The direction of magnetization obtained from these rocks is Dec 244.2; Inc −3.9; $\alpha_{95} = 8.0$; $k = 41.9$, based on nine sites, which corresponds to a palaeopole position at 25.3°S, 313.7°E ($d_p = 4.0$, $d_m = 8.0$). This pole is significantly different from other palaeopole positions of comparable age from Eastern Africa and indicates a separation between the NHT and the ancient eastern margin of Africa of approximately 3000 km at the time of magnetization.

1. Introduction

Nearly all of the crust of the Arabian–Nubian shield (ANS) in northeastern Africa and Arabia was generated during the late Precambrian to early Palaeozoic Pan-African evolution (950–450 Ma) [1]. Detailed structural, petrological and geochemical studies in the region have resulted in the identification of several highly strained zones, containing dismembered ophiolite fragments. These ophiolite belts are believed to separate distinct accreted arc terranes (Fig. 1) and support geodynamic models such as present-day arc/back-arc type collision and terrane accretion for the evolution of the ANS.

Although there is still debate with regard to the direction of subduction and accretion [2], northwestward subduction of oceanic crust under the active North African continental margin, and subsequent accretion of arc-terranes seems to emerge as the most likely scenario for the geodynamic evolution of the eastern margin of North Africa during the late Proterozoic [1]. Geochemical data, although of limited use for quantitative reconstructions of the spatial distribution of lithospheric plates and microplates, have nevertheless been utilized to demonstrate the arc-type character of these terranes and their relative proximity to the northeastern African continental margin [1].

Palaeomagnetic methods have proved a very powerful instrument for the identification and delineation of the geodynamic evolution of com-

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plex accretionary orogens. However, they have, so far, not been successfully applied to the quantification of tectonic processes along the eastern margin of the North African craton during late Proterozoic times. The only relevant palaeomagnetic study from the ANS published so far was on intrusive rocks from the Arabian Peninsula [3] and did not yield any palaeomagnetic results predating the widespread Pan-African plutonism (about 600 Ma). There is, therefore, up to now, no palaeomagnetic information on the relative position of the various terranes prior to accretion.

In this paper we present new geochronological data from zircons (conventional U–Pb, and single grain $^{207}Pb/^{206}Pb$) and single grains of hornblende and biotite ($^{39}Ar/^{40}Ar$). This is combined with new palaeomagnetic data from the Northern Haya Terrane (NHT), a late Precambrian arc terrane from the Red Sea Hills (RSH), northeast Sudan, in order to constrain the geodynamic position of the RSH during the Pan-African evolution and prior to the final accretion to the ancient African craton.

2. The geology of the northern Haya terrane

The Haya terrane (Fig. 1) is one of the volcanic arc terranes that were identified as constituents of the RSH [1]. It is separated by ophiolite suture zones, both to the northwest from the ca. 830 Ma old Gebeit terrane [4], and to the southeast from the ca. 850 Ma old Tokar terrane [1]. The WSW–ENE striking Ashat shear zone [5,6] divides the Haya terrane in a southern part and a northern part, the latter of which is the subject of this study (Fig. 2). The NHT is largely composed of a volcano-sedimentary sequence known as the Nafirdeib Series [7]. The volcanic rocks of this series range in chemical composition from basic to acidic, with basalts and rhyolites being the dominant rock types. The volcanic rocks were classified geochemically as the effusive constituents of an intra-oceanic island arc [4].

The sediments comprise acidic pyroclastics and limestones, the latter recrystallized to impure marbles. Although clastic sediments are not exposed in the area studied, immature sediments,

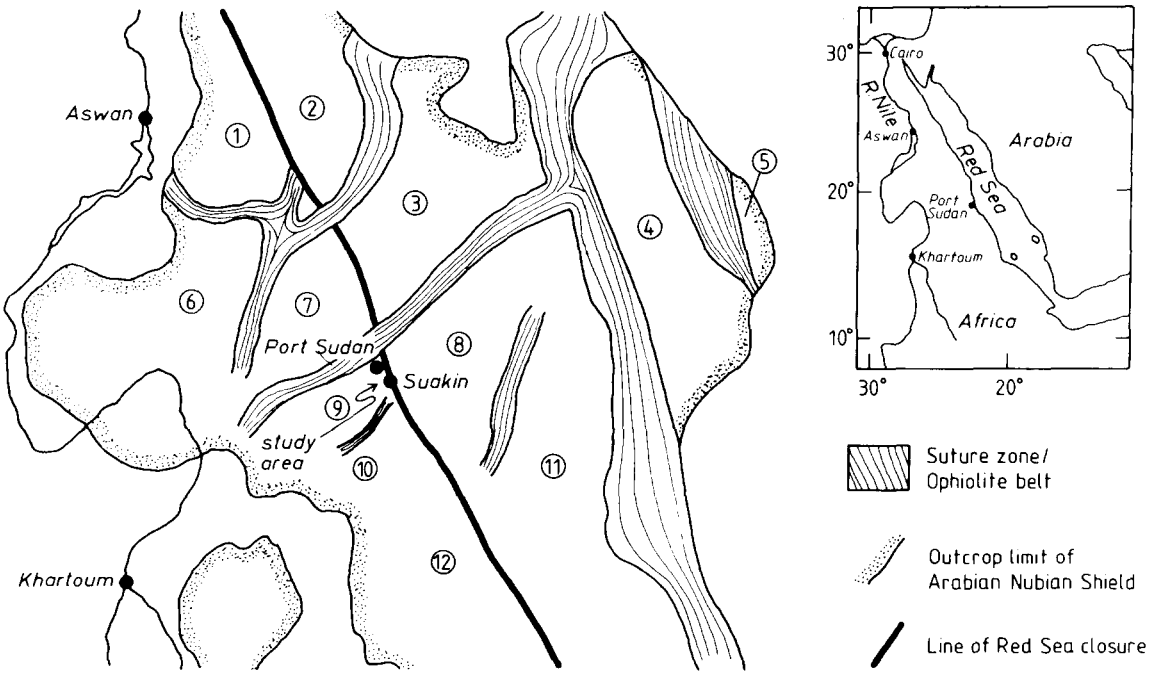


Fig. 1. Geological sketch map of the Arabian–Nubian shield showing major terranes and ophiolite zones, redrawn after Kröner et al. [1]. Numbers indicate terranes: *1* = Gerf terrane; *2* = Midyan terrane; *3* = Hijaz terrane; *4* = Afif terrane; *5* = Ar Rayn terrane; *6* = Gabgaba terrane; *7* = Gebeit terrane; *8* = Taif terrane; *9* = northern Haya terrane; *10* = southern Haya terrane; *11* = Asir terrane; *12* = Tokar terrane.

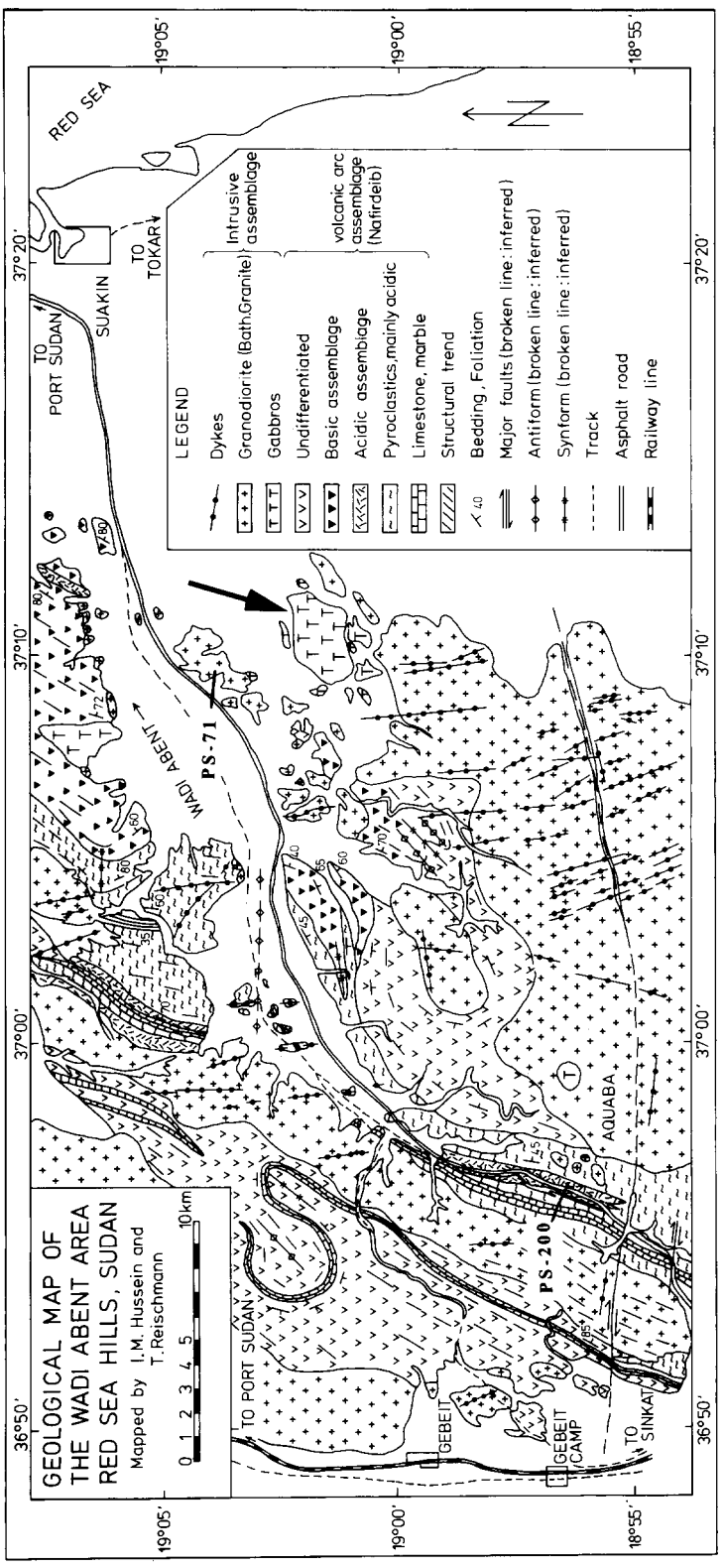


Fig. 2. Geological map of the study area west of Suakin, Red Sea Hills, after Reischmann [4]. Arrow points to the Suakin gabbro. Locations are also indicated for geochronological samples of the Suakin granite (PS-71) and the rhyolites (PS-200).

such as conglomerates and greywackes, have been identified some 50 km to the northwest of the study area. The rocks of the Nafirdeib series are foliated, folded and metamorphosed to the greenschist facies, locally even to amphibolite facies. They were intruded by a suite of syn- to late-tectonic plutons, ranging in composition from gabbro through diorites and granodiorites to granites, all being part of the Batholithic Granite [7] that bears chemical and petrographical I-type characteristics. The compositional variation has been found to reflect a chronological fractionation sequence, with the gabbros being the oldest members, which were subsequently intruded by the more highly differentiated granitoids. Post-tectonic igneous activity is documented by N–S and E–W trending dykes, which mark the end of Precambrian igneous activity in this terrane. The deformation pattern of the region is controlled by E–W trending strike-slip faults and N–S or NNW–SSE striking shear zones [8].

3. Geochronology

Radiometric ages from the NHT are rare. Vail and Hughes [9] reported whole-rock K/Ar ages on gabbroic and dioritic dykes, from ca. 100 km to the south of our study area which range between 663 ± 14 Ma and 574 ± 12 Ma. These dykes are post-tectonic and truncate granitic plutons near Erkowit, thus they reflect a minimum age for the main phase of igneous activity and crust formation of this region. Other geochronological information for the Haya terrane, such as for the Erkowit granite, varies between a Rb/Sr whole-rock isochron age of 815 ± 25 Ma [10] and a single zircon age of 852 ± 13 Ma [6]. The Dahand tonalite, another pluton of the Erkowit region, yields an intrusion age of 871 ± 10 Ma (single zircon evaporation) [6]. These data are in good agreement with geochemical and geochronological data from the neighbouring Gebeit terrane [4], the Tokar terrane [6] in the Red Sea Hills and the Lith terrane [11] in the Arabian shield, and, therefore, they may document major episodes of magmatism in the early Pan-African evolution of this part of the ANS.

3.1 U–Pb and $^{207}Pb/^{206}Pb$ zircon geochronology

Acidic volcanic rocks from the Aquaba Pass and a granodiorite from Wadi Abent, to the west of Suakin (the so-called Suakin granite) were chosen for zircon geochronology to date the formation age of the major rock types studied. The U/Pb system in zircon is believed to preserve the age of rock formation, as, in most cases, it is not reset by regional low-grade metamorphism. Therefore, the zircon dating methods are the most suitable methods for revealing the age of rock formation; other methods, such as Rb/Sr whole-rock dating, are more susceptible to later alteration during metamorphism [6].

The age of emplacement for the Suakin gabbro was not determined directly from the gabbro itself, but from indirect evidence. This is because zircons are not frequent accessory minerals in gabbros and the rare zircons which are present might be xenocrysts and thus not of the same age as the gabbro. Since the gabbro intruded the volcano-sedimentary Nafirdeib Series after it was folded and regionally metamorphosed, the formation age of this rock series provides a upper age limit for the gabbro intrusion. As the gabbro, in turn, is intruded by the Suakin granite, it is, therefore, the earliest recognizable phase of this plutonic suite and predates the Suakin granite emplacement. The ages of the Nafirdeib Series and the Suakin granite thus provide the upper and lower age brackets for the gabbro intrusion. Samples from the Suakin granite (PS-71) contained sufficient quantities of zircon, allowing separation of three zircon fractions. The volcanic rocks of the Nafirdeib Series, however, did not contain significant amounts of zircons at all, and even the most acidic flow-banded rhyolites from the Aquaba Pass (PS-200) yielded only a few small (< 100 μm) zircons. Therefore, only one fraction of zircons could be separated for dating.

The zircons for U–Pb analyses were dissolved in sealed teflon bombs at a temperature of 200°C for 1 week using a mixture of concentrated HF–HNO_3. The spike applied was highly enriched in ^{233}U and ^{202}Pb. After chemical separation, Pb was loaded with silica gel on a Re single filament. U was loaded with graphite also on a Re single filament. Isotope ratios were measured in the Max-Planck-Institut für Chemie in Mainz using a Finnigan MAT 261 mass spectrometer with a multi Faraday cup configuration for Pb and a secondary electron multiplier (SEM) for U. Data reduction includes corrections for fractionation,

TABLE 1

U–Pb analyses of zircons from Suakin granite PS-71 and Aquaba rhyolite PS-200

Fraction	Weight (mg)	U (ppm)	Pb [2] (ppm)	Observed isotope ratios [2]			Corrected isotope ratios [3]			Ages (Ma)		
				$^{206}Pb/^{204}Pb$	$^{207}Pb/^{206}Pb$	$^{208}Pb/^{206}Pb$	$^{206}Pb/^{238}U$	$^{207}Pb/^{235}U$	$^{207}Pb/^{206}Pb$	$^{206}Pb/^{238}U$	$^{207}Pb/^{235}U$	$^{207}Pb/^{206}Pb$
PS-71A	0.138	269	31	925	0.083333	0.220542	0.100153	0.937396	0.067883	615 ± 6	672 ± 10	865 ± 25
PS-71B	0.241	138	16	687	0.088261	0.194621	0.102811	0.956021	0.067442	631 ± 5	681 ± 7	851 ± 17
PS-71C	0.395	179	20	1304	0.078429	0.171074	0.103798	0.965278	0.067447	637 ± 5	686 ± 8	852 ± 20
PS-200	0.462	44	8	250	0.125905	0.248881	0.139256	1.315088	0.068596	840 ± 6	853 ± 7	887 ± 20

[1] Total Pb.

[2] Corrected for mass fractionation and spike.

[3] Corrected for common Pb.

PS-71A: White, transparent, long, euhedral, 70–200 μm.

PS-71B: White, transparent, short, euhedral, 70–200 μm.

PS-71C: Colourless, clear, short, euhedral, 70–200 μm.

PS-200: Pink, clear, mixed sizes with all zircons found.

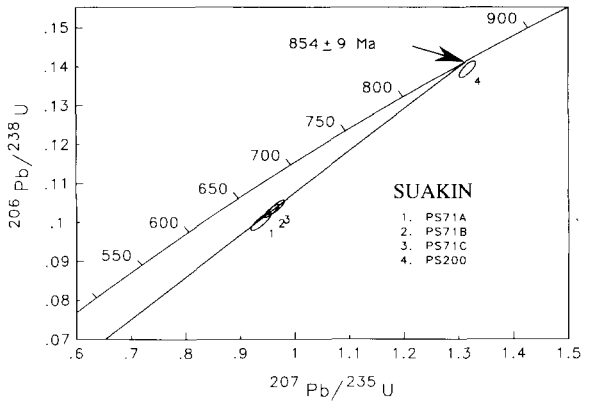


Fig. 3. Concordia diagram showing the data points for three zircon fractions of Suakin granite PS-71 and one for the Aquaba rhyolite PS-200. The rhyolite data are not included in the regression.

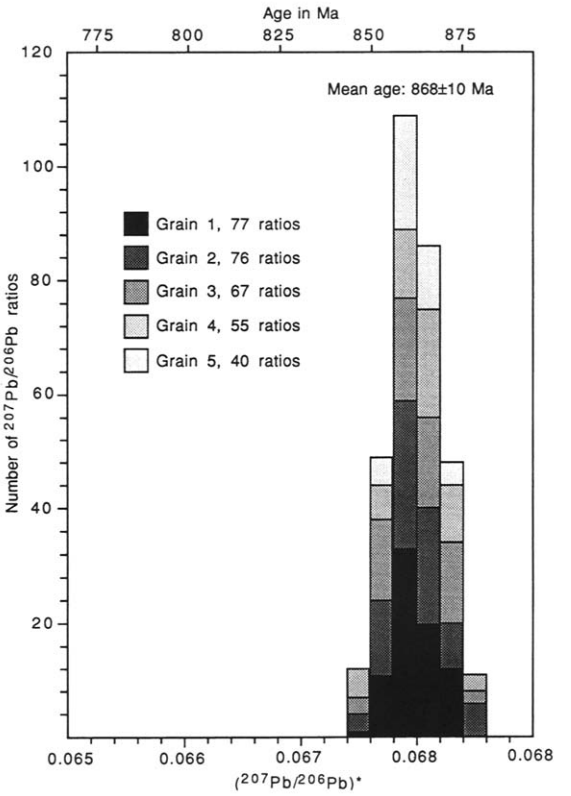


Fig. 4. Histograms showing the results of the $^{207}Pb/^{206}Pb$ single grain zircon analyses of the Aquaba rhyolite PS-200. The spectrum has been integrated from 315 ratios.

blank and common Pb after Stacey and Kramers [12]. Ages are calculated by linear regression after York [13], and errors are quoted at the 2σ level. $^{207}Pb/^{206}Pb$ single zircon ages were obtained using Re filaments and the SEM. This method is described in Kober [14] and Kröner et al. [6].

The analytical data are listed in Tables 1 and 2. The U/Pb data are plotted in the concordia diagram shown in Fig. 3. The single zircon $^{207}Pb/^{206}Pb$ data are plotted in a histogram in Fig. 4. The data for the Suakin granite (PS-71) were regressed through the origin, yielding an age of 854 ± 9 Ma (MSWD = 0.8) which we interpret as the age of intrusion. Regression through the origin was chosen because regression of the data points alone yields a negative lower intercept. The data point for the rhyolite (PS-200) with a $^{207}Pb/^{206}Pb$ age of 887 ± 20 Ma is close to, but significantly off, the regression line. The uncertainty of 20 Ma is caused by the very low U content of 44 ppm (Table 1) and the resulting low content of radiogenic Pb, which is also evident from the low $^{206}Pb/^{204}Pb$ ratio of 250. The single

TABLE 2

Isotopic data from single grain zircon evaporation of rhyolite PS 200

Grain	Morphology	Mass scans [1]	Evaporation temp. (°C)	Mean $^{207}Pb/^{206}Pb$ ratio [2] and 1σ error	$^{207}Pb/^{206}Pb$ age and 2σ error
1	stubby, clear, euhedral	77	1560	0.06798 ± 17	868 ± 10
2	stubby, clear, euhedral	76	1510	0.06799 ± 24	868 ± 14
3	stubby, clear, euhedral	67	1520	0.06799 ± 27	868 ± 16
4	long, clear, euhedral	55	1540	0.06802 ± 25	869 ± 16
5	long, clear, euhedral	40	1550	0.06798 ± 15	868 ± 10
1–5		315		0.06798 ± 15	868 ± 10

[1] Number of $^{207}Pb/^{206}Pb$ ratios evaluated for age assessment.

[2] Observed mean ratio corrected for non-radiogenic Pb where necessary. Errors based on uncertainties in counting statistics.

zircon $^{207}Pb/^{206}Pb$ age of 868 ± 5 Ma for the rhyolites is also in good agreement with the conventional $^{207}Pb/^{206}Pb$ age. Therefore, the time of volcanic eruption can be bracketed between ca. 887 and 868 Ma. From field observation it is evident that gabbro emplacement postdates the folding and low to medium grade regional metamorphism of the Nafirdeib Series, and because of the genetic relationship is likely to be close to the emplacement age of the Suakin granite at 854 Ma. Our data are similar to those for the Erkowit region some 100 km farther south [6] and again prove considerable crustal growth during the 900–850 Ma interval in the Pan-African evolution.

3.2. *Hornblende and biotite single grain $^{40}Ar/^{39}Ar$ dating*

Hornblende and biotite grains from the Suakin Gabbro were obtained from palaeomagnetic cores from site SGG (Table 3) and were dated by the single-grain step-heating technique using the VG Micromass 1200S mass spectrometer at the University of Toronto [15]. The mineral grains and three grains of hb3gr (1071 Ma, Zartman [16]) were irradiated on aluminum trays for 192 MWhr at the McMaster University Nuclear Reactor. Each standard was fused in a single step, and the irridation parameters (J) from the three grains were averaged to determine a J for the tray. The mineral grains were step-heated using a continuous argon-ion laser with the beam size of sufficient diameter to cover the grain completely and ensure uniform heating. The temperature of the grain during heating was monitored using an infra-red microscope.

One biotite grain was step-heated and 14 fractions were collected (Fig. 5). The biotite age spectrum exhibits a classic "stair-step" pattern indicative of diffusional argon loss. A plateau is reached for the last 60% of gas release with an age of 782.5 ± 6.4 Ma (2σ error). A single loss diffusional model also yields a formation age of about 784 Ma, with 15% gas loss at about 596 Ma.

Five hornblende grains were step-heated with 9–14 fractions collected per grain. Sample SGG–1 showed the least disturbed spectrum, with a plateau age of 840.6 ± 4.2 Ma (Fig. 5). The plateau consists of four adjacent fractions between 930°C and 1020°C, which comprise 50% of the gas release. Higher temperature fractions have an age which is the same as the plateau age. However, there is a dip in the age spectrum at 1050°C which prevents us from considering all fractions of gas release above 930°C as a proper plateau.

All hornblende grains show some evidence of argon loss, both at low temperature release and at about 80% ^{39}Ar release (Fig. 6). Four hornblende grains have ages which scatter between the SGG-1 hornblende and the biotite ages. These hornblendes are considered as altered grains. Of these, SGG-4 has a single fraction with an age of 836.9 ± 8.8 Ma, identical to the SGG-l plateau age. The Ca/K spectra for the altered hornblendes also shows considerable scatter between the Ca/K for SGG-1 hornblende and the biotite spectra.

TABLE 3

Summary of Suakin gabbro $^{40}Ar/^{39}Ar$ single grain step heating

Grain	Integrated age (Ma)	Integrated Ca/K	Age of oldest fraction (Ma)	Plateau age (Ma)	^{39}Ar in plateau (%)
Hornblende SGG					
1	824.1 ± 2.4	14.78 ± 0.04	842.4 ± 4.4	840.6 ± 4.2	50.7
2	776.6 ± 2.1	3.17 ± 0.01	819.8 ± 4.0		
3	807.7 ± 2.5	8.22 ± 0.02	828.2 ± 6.2		
4	814.0 ± 8.4	10.69 ± 0.04	836.9 ± 6.0		
5	754.3 ± 6.6	10.53 ± 0.03	813.5 ± 4.0		
Biotite SGG					
1	757.6 ± 6.1	0.04 ± 0.00	785.9 ± 2.9	782.5 ± 6.4	57.5

Errors quoted at 2σ level and include all analytical errors. Integrated and plateau age errors include uncertainties in estimating the irradiation parameter, J.

Figure 7 is a plot of individual fraction ages for the five hornblende and one biotite sample versus the Ca/K ratio for the fraction. It can be seen that there is a remarkable correlation between the Ca/K content and the age of the fraction for the hornblende, scattering between the biotite with Ca/K = 0 and the hornblende SGG-1 with Ca/K = 17. Thus, each of the fractions of the altered hornblende can be thought of as a mixing of two different argon and calcium reservoirs. The first is a hornblende phase with an age of about 841 Ma and a high Ca/K, while the second is a biotite phase with an age of 783 and no calcium. We interpret these two phases as representing varying degrees of biotite intergrowth within individual hornblende grains that had an original age of about 841 Ma. The biotite age of

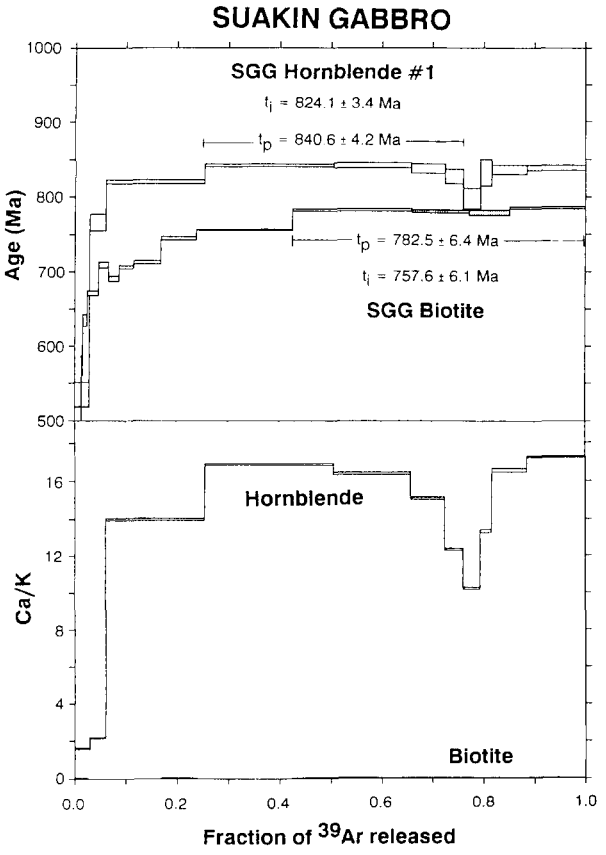


Fig. 5. Biotite and least altered hornblende age and Ca/K spectra from Suakin gabbro sample SGG. Height of boxes is the 1σ error on the age of each fraction. The plateau (t_p) and integrated (t_i) ages are shown with their 2σ errors.

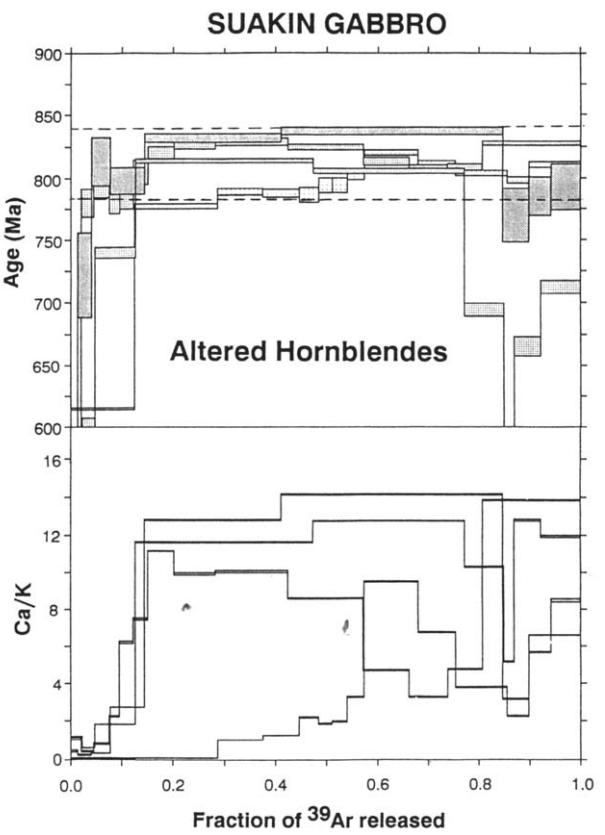


Fig. 6. Age and Ca/K spectra of the four more-altered hornblende grains from sample SGG. Horizontal dashed lines denote the plateau ages of the least altered hornblende and biotite samples (from Fig. 5).

783 Ma could either be interpreted as the time of closure of the biotite system after slow cooling for ca. 60 Ma or as an indication of a biotite alteration event at that time. The second alternative is more likely because retrograde reactions of hornblende to biotite and actinolite and later chloritisation of these three minerals could be detected in thin sections of the gabbro. Although these reaction products were not identified under the binocular microscope in the minerals analyzed, we cannot exclude the possibility that minute impurities were still hidden in the opaque hornblende and biotite grains.

Despite the complications caused by retrograde reactions, we can conclude from the single-grain dating of hornblende and biotite, that the gabbro passed through the argon closure temperature of hornblende (about 550°C) at 840.6 ±

4.2 Ma. The similarity between the magnetite blocking temperature (ca. 570°C) and the hornblende argon closure temperature allows us to assign the hornblende $^{40}Ar/^{39}Ar$ age as the best estimate of the age of magnetization of the Suakin gabbro. The gabbro either cooled slowly over the next 60 Ma to the biotite blocking temperature or was reheated in a biotite-forming event at 783 Ma, which caused alteration of some hornblende grains. However, there is no documented geological event known from the NHT at that time. Closure of the Amur-Nakasib suture zone, which borders the NHT to the northwest, thus docking with the Gebeit terrane, is assumed to have occurred at ca. 700 Ma [17]. The motion along the major N–S striking shear zones was even later, and thus did not cause any event documented by the biotite samples. However, from ca. 150 km to the south in the Southern Haya Terrane (SHT), Vail [18] reported K/Ar biotite ages of up to 786 Ma. In the Asir terrane, which is the eastern continuation of the NHT into Saudi Arabia, igneous activity from ca. 780 Ma is documented [19] and igneous activity is also known from the northern ANS [20]. Although not yet recognized in the NHT, we might expect a geological event responsible for the 783 Ma biotite ages.

4. Palaeomagnetism

4.1 Palaeomagnetic methods

The sediments and volcanic rocks of the Nafirdeib Series and the stratigraphically younger gabbros and granitoids were sampled at 44 sites. Samples were either taken as oriented hand samples or cored in the field using a portable gas-powered drill with a diamond drill bit and oriented using a standard sun compass. Standard palaeomagnetic specimens (2.2×2.5 cm diameter) were prepared from oriented core and/or hand samples, and their natural remanent magnetization (NRM) was measured on a three-axis ScT cryogenic magnetometer at the ETH, Zürich. Thermal demagnetization experiments were carried out in air using a magnetically shielded (residual magnetic field < 5 nT) Schönstedt TSD-1 furnace. Alternating field (AF) demagnetization was performed using a 3-axis tumbler within a large solenoid, capable of achieving peak alternating fields at 200 mT. A set of Helmholtz coils surrounding the AF device was used to reduce the intensity of the direct field to about 10 nT during the demagnetization process.

The results of the stepwise AF or thermal

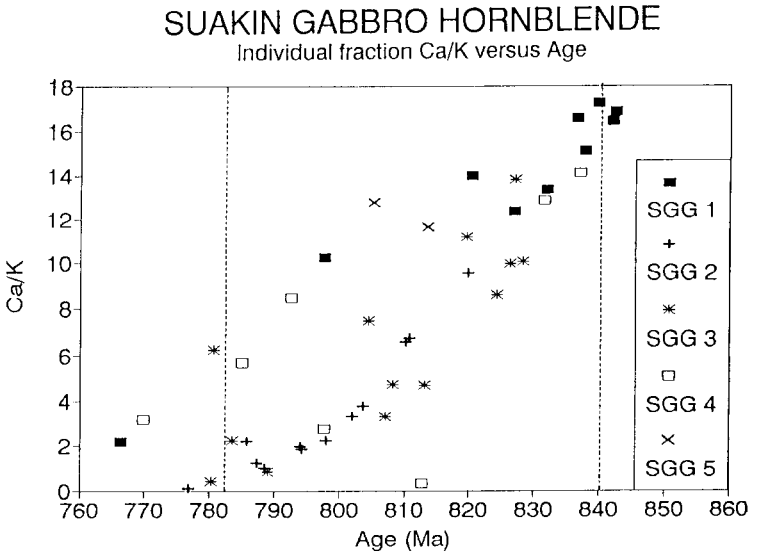


Fig. 7. Plot of the Ca/K value versus the apparent age of each fraction from the five hornblende grains. Vertical dashed lines denote the biotite and the SGG 1 hornblende plateau ages (from Fig. 5). There appears to be a strong correlation between the Ca/K and the age of a fraction, implying that each fraction is actually a mixture between a low calcium, young end member (biotite) and a high calcium, old end member (hornblende).

demagnetization experiments for individual specimens are plotted on equal area projections and orthogonal vector diagrams [21]. Linear and planar demagnetization trajectories were identified by visual inspection and calculated using the Kirschvink algorithm [22]. Only lines and/or planes containing 4 or more successive demagnetization steps were judged reliable and used to compute site mean directions.

4.2. *Palaeomagnetic results*

Detailed AF and thermal demagnetization studies on selected samples from the Nafirdeib Series (dolomites, acidic and basic volcanic rocks) either did not reveal any consistent direction of the NRM at all, or, as for the syn- to lateorogenic granites from Suakin and Erkowit, only yielded directions, which were indistinguishable from the direction of the present dipole field at the sample location. These samples were, therefore, interpreted as reflecting complete recent and/or Pleistocene magnetic overprints. All samples revealing inconsistent results or present field directions are not considered any further in this paper.

Thus, most of the data had to be rejected with the exception of the samples (9 sites) from the Suakin gabbro (Fig. 2) which displayed surprisingly simple magnetic behaviour during thermal (Fig. 8a and c) and AF demagnetization (Fig. 8b and d). The NRM of the majority of the samples is dominated by a very strong viscous(?) component with randomly distributed directions. Thermal demagnetization up to about 200°C, or AF-demagnetization with peak fields of about 20 mT, was generally sufficient to remove this component completely, which in some cases accounted for more than 80% of the total intensity of the

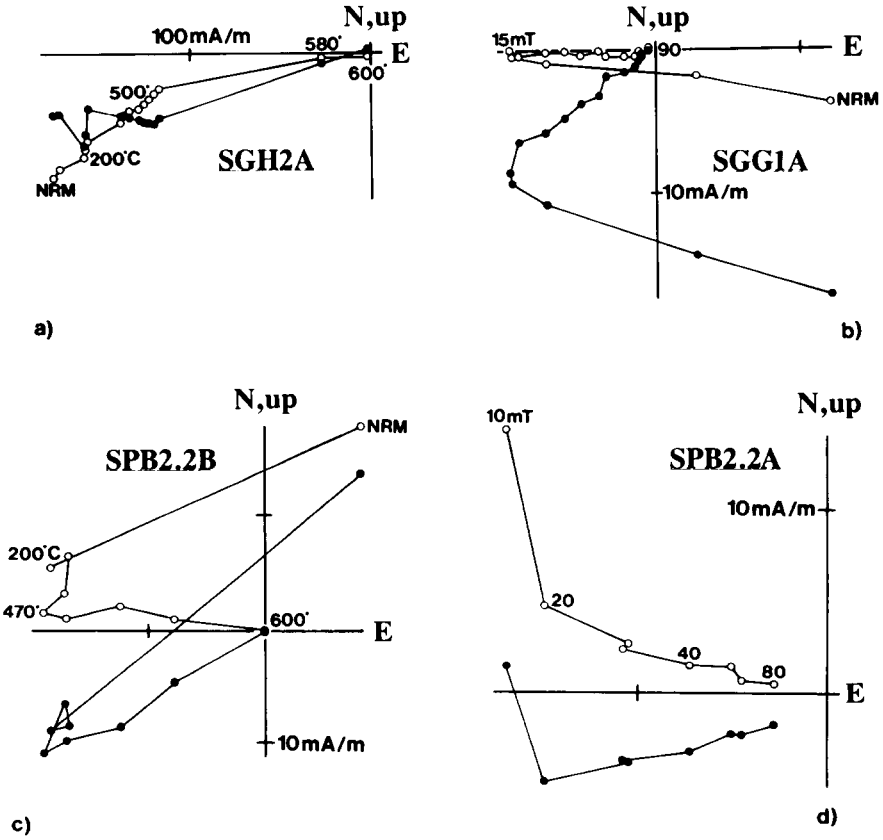


Fig. 8. Orthogonal thermal (a, c) and AF (b, d) demagnetization diagrams of representative samples from the Suakin gabbro. ● = projections of the demagnetization vector on the horizontal plane; ○ = projections on the east–west vertical plane.

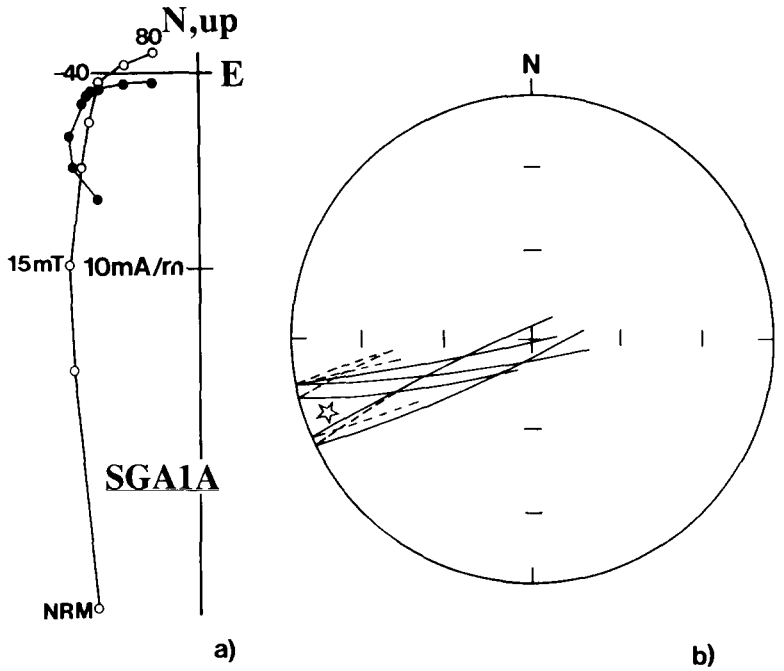


Fig. 9. Orthogonal AF demagnetization diagram of a sample (a) from site SGA and (b) remagnetization circles from other samples from site SGA. The common mean direction (Dec/Inc: 250.5/−10.4), computed after McFadden and McElhinny [23] is marked by the star. Symbols as in Fig. 8.

NRM. Upon removal of this low coercivity, low blocking temperature component, southwesterly and very shallow characteristic directions (ChRM) can easily be identified as linear segments trending towards the origin of the orthogonal projection.

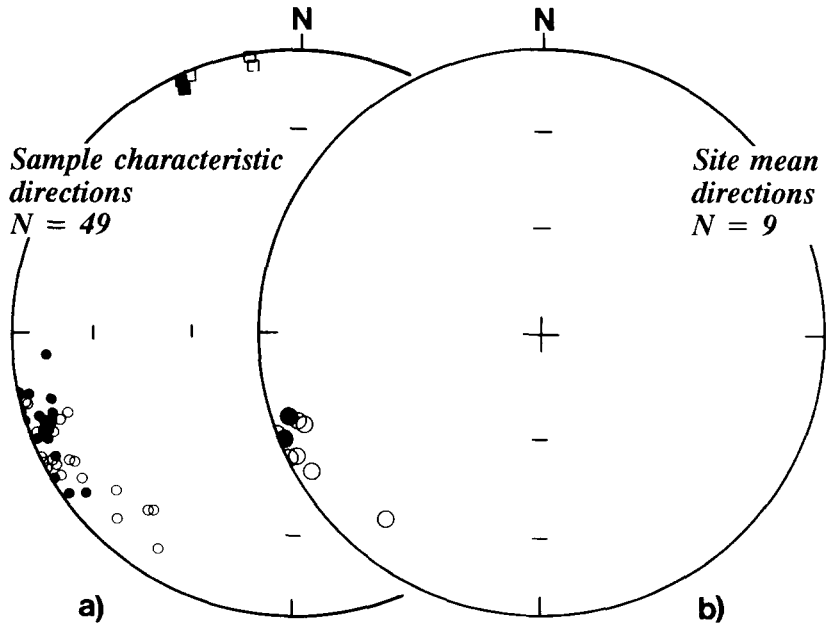


Fig. 10. Equal area projection (a) of characteristic directions and (b) of site mean directions. White = projections on the upper hemisphere, black = projection on lower hemisphere; square = poles to great circles.

Only samples from site SGA did not display such simple magnetic behaviour during thermal and AF demagnetization (Fig. 9a). Overlapping blocking temperature and coercivity spectra of two components resulted in curved demagnetization trajectories (Fig. 9b) trending towards the southwest with shallow inclinations. Applying the McFadden and McElhinny [23] algorithm for combining remagnetization circles and using sector constraints of 180/80 and 270/−20 along individual great circles gives a common mean direction for samples from site SGA, which is indistinguishable from the mean directions based on stable endpoints obtained from other sites (Fig. 10 and Table 4). This observation lends additional support to our assumption that the ChRMs identified in samples from the remaining sites and listed in Table 4 have indeed been properly isolated.

The blocking temperature (approximately 570°C) in connection with the observed coercivity characteristics (approximately 90 mT) of the ChRM in all samples from Suakin are diagnostic for (single domain?) magnetite. Microprobe analysis of representative samples also reveals the presence of pure magnetite as the exclusive magnetic phase in these samples. Since the magnetite blocking temperature is well approximated by the $^{39}Ar/^{40}Ar$ closure temperature of hornblende, the latter age is representative for the time the ChRM was achieved. The site mean direction (Dec 244.2, Inc −3.9, $\alpha 95 = 8.0$, $k = 41.9$, 9 sites) carried by magnetite is, therefore, taken as the ChRM of the Suakin Gabbro, and it corresponds to a palaeopole position at 25.3°S, 313.7°E ($d_p = 4.0$, $d_m = 8.0$) at 841 Ma.

5. Discussion and conclusion

Van der Voo and Meert [24] have shown that the available palaeomagnetic data set does not allow the construction of a single APWP for

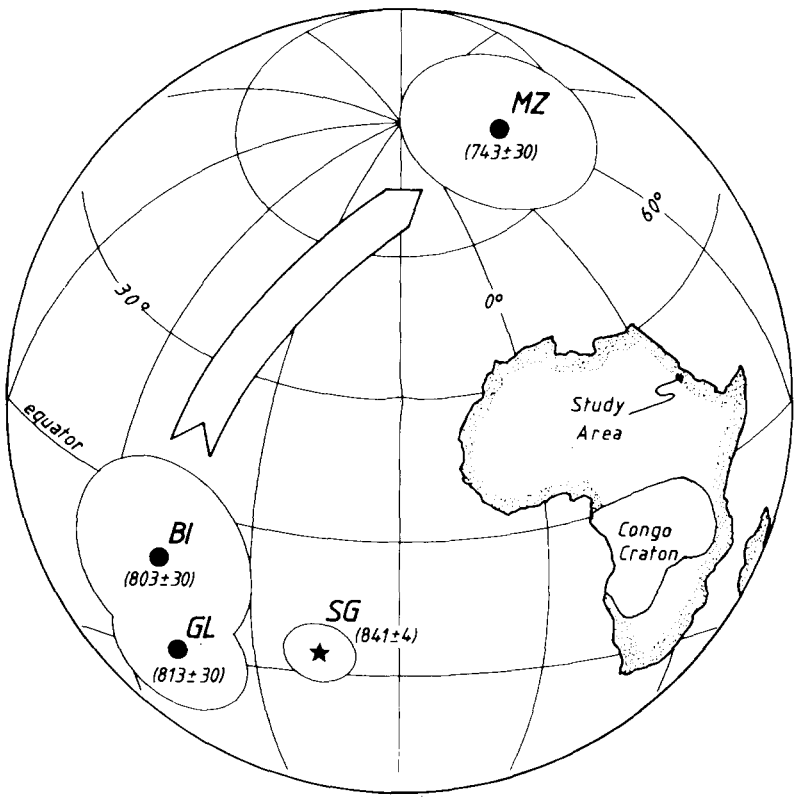


Fig. 11. Palaeopole position for the Suakin gabbro (star). Also shown is the Apparent Polar Wander Path (south poles) for the Congo craton for the 813–743 Ma interval, after Van der Voo and Meert [24]. Labels of reference poles as in Table 5, magnetization ages are given in brackets.

TABLE 4

Summary of site-mean directions from the Suakin gabbro (19.033°N, 37.166°E)

Site	n/N	Dec	Inc	k	α_{95}
SGA *	5/5	250.5	−10.4	44.4	16.9
SGB	4/4	243.9	+0.1	46.6	13.6
SGC	5/5	220.1	−16.9	113.1	7.2
SGD	7/7	249.4	−0.8	110.3	5.8
SGE	6/6	239.1	−7.8	286.5	4.0
SGF	4/4	249.3	−12.5	63.8	11.6
SGG	6/6	247.7	+3.4	169.3	5.2
SGH	7/7	251.7	+8.3	79.5	6.8
SPB	5/5	243.3	−5.4	53.6	10.5
Mean	9/9	244.2	−3.9	41.9	8.0
Palaeopole position					
		Lat (°S)	Long (°E)	d_p	d_m
		25.3	313.7	4.0	8.0

n/N = the ratio of samples used for the composition of the mean to the samples demagnetized, statistics after Fisher [26].
k = precision parameter of the mean direction.
α_{95} = semi-angle of the cone of confidence about the mean direction at the 95% confidence level.
Lat and Long = latitude and longitude of the palaeomagnetic pole position.
d_p, d_m = semi-minor and semi-major axis of the polar error ellipse at the 95% confidence level.
* The site mean direction for site SGA has been computed by combining great circles [23].

Africa in the late Proterozoic. However, it is possible to define rather short APWP segments for the late Proterozoic for three major African cratons, namely: the Congo craton, the Kalahari craton, and the West African craton [24].

Since the Congo craton is the African craton with a reliable APWP segment for the 813–743 Ma age interval, which is closest to the Suakin Gabbro (SG) magnetization age of 841 ± 4 Ma, palaeopoles from this APWP segment have been chosen as reference. We selected poles from the Congo craton, meeting the first Van der Voo [25] criterion which requires the rock age to be determined within 4% of the numerical age, or with a precision better than 40 Ma. These reference poles have been plotted in Fig. 11 and are listed in Table 5, together with the assigned Van der Voo quality index, Q.

The magnetization ages of the two reliable palaeopoles from the Congo craton (BI and GL, Table 5) are 813 ± 30 and 806 ± 30 Ma, respectively. These pole positions differ significantly from the SG pole (25.3°S, 313.7°E, $d_p = 4.0$, $d_m = 8.0$). The SG pole is displaced some 30° to the east of the APWP segment of the Congo craton (Fig. 11), indicating a substantial separation of ca. 3000 km between the latter and the NHT. This interpretation, however, is based on the assumption of a contemporaneous magnetization, which is acceptable taking into account the uncertainties of the magnetization ages of the reference poles. According to this interpretation the NHT must have joined the Congo craton after ~ 810 Ma.

Assuming that the the time of magnetization for the SG and the two reference poles GL and BI differs by ca. 30 Ma, alternative explanations ought to be considered. Since the shape of the APWP for the Congo craton prior to 813 Ma is unknown [24], coherence between the Congo craton and the NHT cannot be ruled out completely. An age difference of about 30 Ma would correspond to a plate movement of about 3000 km at a velocity of 10 cm/a, which is a reasonable value for modern plate motion. Although the

TABLE 5

Compilation of Late Proterozoic pole positions for the Congo Craton fullfilling the age criterion after Van der Voo [25]

Abbr. Formation, Location	Age range (Ma)	Lat.	Long.	α_{95}	k	Q	Reference
Mz Mbozi Complex, Tanzania	713–773	72.2°N	67.7°E	18.0	18.5	2	[27]
GL Gagwe Lavas, Tanzania	783–843	29.0°S	283.0°E	13.0	5.3	3	[28]
BI Bukoba Intrusives, Tanzania	776–836	11.0°S	281.0°E	19.0	4.7	4	[28]
SG Suakin Gabbro, Sudan	837–845	25.3°S	313.7°E	6.7	59.3	5	this study

α_{95} = semi-angle of cone of confidence about the palaeopole position at the 95% probability level.
Lat., Long. = latitude and longitude of the palaeomagnetic pole.
k = Fisherian precision parameter associated with the means.
Q = quality index after Van der Voo [25].
The age range gives an estimate of the likely ages of the rocks studied and the age of magnetization.

geochronological and palaeomagnetic data of this study are precise and accurate, the comparison with data from late Proterozoic APWP segments for Africa does not allow a sound conclusion as to the position of the NHT relative to the African cratons. It is evident from this discussion, that the difficulties in the interpretation of the palaeomagnetic data for the SG are caused by the scarcity of high-quality palaeomagnetic data and the associated uncertainties of the APWP of Africa in the late Proterozoic. Better palaeomagnetic and geochronological data from the African cratons are required for a reliable geodynamic interpretation.

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